

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 15-300-4
TERMINAL LUGS FOR CABLES UP TO 30 kV($V_m=36KV$)	Date: 28-4-2026

EDMS 15-300-4

SPECIFICATION

FOR

TERMINAL LUGS

FOR CABLES UP TO 30 kV($V_m=36KV$)

(ALUMINUM / COPPER / BI-METALLIC)

Issue: April-2026 / Rev- 4

This revision contains option items that may be requested by distribution company before bidding

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1. SCOPE

This specification describes the minimum technical requirements for the design, engineering, manufacture, testing, supply and delivery of lugs for connecting cables/conductors to distribution points for cables up to 18/30 kV ($V_m = 36$ Kv). The required lugs shall be suitable for use in both indoor and outdoor installations in the distribution system of Egyptian Electricity Distribution Companies (EDCs).

2. APPLICABLE CODES and STANDARDS

This specification shall be read in conjunction with the latest revisions of the following EDMS specifications:

EDMS 03-300	L.V Cables, Aluminum or copper 0.6/1 KV.
EDMS 03-400	M.V Cables 12/20 KV and 18/30 KV

The equipment/material covered in this specification shall comply with the latest edition/amendment of the following codes and standards. Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply.

IEC 60228	Conductors for Insulated Cables.
IEC 61238-1-1	Compression and Mechanical Connectors for Power Cables for Rated Voltages up to 1 kV ($U_m = 1.2$ kV) Part: 1–Test Methods and Requirements.
IEC 61238-1-3	Compression and Mechanical Connectors for Power Cables for Rated Voltages above 1 kv. up to 30 kV ($U_m = 36$ kV) Part: 1–Test Methods and Requirements.
DIN EN 13600	Copper and copper alloys - Seamless copper tubes for electrical purposes.
DIN EN 755	Aluminum and aluminum alloys - Extruded rod/bar, tube and profiles.
DIN EN 755-7	Manufacturing tolerances
DIN 46329	Cable Lugs for Compression, Ring Type, for Aluminum Conductors
DIN 46235	Cable Lugs for Compression, Cover Plate Type, for Copper Conductors
ISO 2859	Sampling procedures for inspection by attributes.
ASTM B544	Standard Specification for Electrodeposited Coatings of Tin.
NEMA/ANSI CC1	Electric Power Connection for Substations.

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3. SERVICE CONDITIONS

Lugs are required to have a good electrical and mechanical performance under the following local service conditions:

Ambient temperature	-5°C to +45°C (50°C is an option) according to --EDC
Maximum relative humidity	95 %
Environmental conditions	Humid tropical climate with heavily polluted atmosphere
Maximum Rainfall	250 mm
Altitude	Up to 1000 m above sea-level

4. DESIGN and CONSTRUCTION REQUIREMENTS

4.1. General:

- All lugs intended for electricity distribution or industrial networks shall be type tested according to IEC 61238-1 Class-A in which they can be subjected to short-circuit of relatively high intensity and duration.
- All lugs shall meet or exceed the performance of conductor in all respects matching to conductor outer diameter, resistance, short circuit, and cross section area.
- Compression type lugs shall be compressed by hexagonal dies.
- Manufacturer's drawings shall show the outline of the lugs together with all pertinent dimensions.
- Manufacturing tolerances shall be as per DIN 755-7, and DIN 13600 for Aluminum and copper respectively.
- The tolerances in lug dimensions shall not exceed the value given in the attached drawings and tables.
- The maximum allowable decrease in weight is 5% of the specified weight.
- The recommended die is the Hexagonal type, with hudrolic crimping tool.
- The crimping area of the compression lugs should not be less than 70% of the lug barrel length.
- Shear bolt mechanical lugs shall be range-taking type suitable for aluminum and copper conductors, with tin-plated aluminum alloy bodies, appropriate number of predetermined shear-off bolts operable by standard hand tools, and removable centering inserts for smaller conductor sizes, providing a reliable mechanical and electrical connection without crimping tools and fully tested and certified in accordance with IEC 61238-1 Class A

4.2. Materials:

- Compression lugs shall be made entirely from copper (Copper lugs), entirely from aluminum (Aluminum lugs) or combination of both metals (Bimetallic lugs) as specified hereafter according to the request of EDC.

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2. All copper parts shall be made of electrolytic tough pitch copper (Cu ETP) of purity not less than 99.9% as per EN 13600. The minimum acceptable conductivity of copper parts is 96.6 % IACS.
3. All aluminum parts shall be made of alloy EN-AW-1050A of purity not less than 99.5% in accordance with DIN EN 755-7. The minimum acceptable conductivity of aluminum parts is 58% IACS.
4. The shear bolt lugs body shall be made of tin-plated aluminum alloy (with minimum conductivity 36% IACS) to be suitable for both copper and aluminum cable conductors

4.2.1. Compression Copper Lugs:

- All copper lugs shall have tin plated surface of thickness 5 μm to protect against corrosion in accordance with DIN 46235.

4.2.2. Compression aluminum Lugs:

- All aluminum lugs shall be made of seamless extruded tubes in accordance with DIN 46329.

4.2.3. Compression bi-metallic Lugs:

- Bi-metallic lugs can be delivered in either friction or insert types according to _EDC requirements.
- Friction type consists of aluminum barrel (seamless extruded tube) jointed with copper palm through friction stir welding.
- Insert type consists of aluminum body (seamless extruded tube) with copper ring inserted in the aluminum palm.

4.2.4. Compression dead-break Elbow Lugs:

- Dead-break elbow lugs should be made by one piece of seamless Aluminum alloy tubes to comply the requirements of EDMS 13-101 and EDMS 13-201.
- Dead-break elbow lugs shall have tin plated surface of thickness 15 μm at least to protect against corrosion.

4.2.5. Shear bolted lugs for terminations and separable connectors:

- Tinned alloy shear bolt mechanical lugs shall be suitable for the cable sizes indicated in the relevant drawings.
- Shear bolts shall be designed to shear within their threaded holes at the specified torque, ensuring no sharp edges, protrusions, or other deformities remain on either the internal or external surfaces of the lug.
- All shear-bolted lugs shall have a tin-plated surface with a thickness of 15:20 μm for corrosion protection.

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4.3. Construction:

- The construction details and dimensions are given in the attached drawings and tables, which form part of this specification.
- All lugs shall have smooth bright surface without any form of nicks, deformations, and sharp edges.
- The barrel of aluminum, bimetallic, and shear bolt lugs shall be filled with joint compound (Contact grease with corundum – nonferrous heavy metal inhibitor). The joint compound shall maintain its characteristics up to 90 °C.
- The barrel of aluminum, bimetallic, and shear bolt lugs shall be capped with an easy to be removed plastic plug.

5. TESTING

- Terminal lugs shall be tested in accordance to the latest relevant standards and this specification. It shall be the responsibility of the manufacture to perform or to have performed all the tests specified.

5.1. Type Tests

- Type tests had to be performed by accredited laboratories, and shall include:
 - A. Short-circuit and heat cycling tests (IEC 61238-1)
 - B. Chemical composition of copper and aluminum (DIN 13600 and DIN 755)
 - C. Tensile tests prescribed in IEC 61238-1.

Lugs shall be able to pass the tensile tests prescribed in IEC 61238-1 and the tensile force for each category shall be as follows:

Class	Conductor Material	Nominal cross-sectional area A (mm ²)	Tensile force (N)
Class 1	Aluminum	≤ 500	$40 \times A$
		≥ 630	20000
	Copper	≤ 300	$60 \times A$
		≥ 400	20000

5.2. Routine Tests

- Routine tests shall be carried out on the charge of the contractor in the presence of EDC representative at the factory. The routine tests shall include:
 - Conductivity test to verify the requirement of clause 4.2 of this specification.
 - Uniformity of the final product with the sample previously attached to the offer.
 - Measurements of dimensions and weights to ensure the compliance with the attached drawings and tables.

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- Sample size: Test specimens shall be selected at random from each inspection lot in accordance to ISO 2859-1. The number of selected samples shall be as follows:

Lot Size	Sample Size	Lot Size	Sample Size
100 pcs. or less	2 pcs.	1001 to 5000 pcs.	25 pcs.
101 to 500 pcs.	6 pcs.	5001 to 20000 pcs.	40 pcs.
501 to 1000 pcs.	12 pcs.	20001 pcs. to over	60 pcs.

- In case of failure of test sample to comply with routine tests, the EDC reserves the right to disqualify the contractor or postpone the delivery until succeeding all the tests. In this case, any delay of supplying shall be on the charge and responsibility of the contractor.

6. MARKING

Lug identification shall be written indelibly way included the following data:

- Manufacturer name (shall be marked to where it will not be affected or erased by crimping),
- Conductor size, & die size,
- Number of crimping, crimping positions (narrow die and wide die) shall be clearly indicated on the outside surface in accordance with the attached tables.
- symbol of conductor materials (AL/CU)

7. PACKING

Packing and delivery shall generally be in suitable carton boxes of maximum 15 kg for each. Each box shall bear the following information legibly printed in a visible place:

- Purchase order number / Tender number
- Lugs catalogue number
- Manufacturer's name
- Year of manufacture
- Quantity of pcs. in the box
- Gross weight in kilograms (kg)
- Position of slinging points and other relevant handling instructions

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8. GUARANTEE

- The supplier shall guarantee the products against all defects arising out of faulty design or manufacturing defects or defective material for a period of two (2) years from the date of delivery.
- The supplier shall guarantee the uniformity of the products delivered with the approved samples.

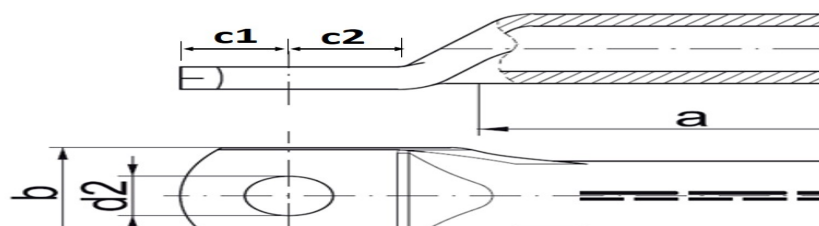
9. SUBMITTALS

- Technical data and original catalogues shall be submitted. The submitted data shall include manufacturer CAD drawings for each of the items offered.
- One specimen of each offered type shall be submitted with the offer for evaluation prior to issuance of the purchase order. Dimensional verification, indelible markings and finishing shall be checked.
- The tenderer must fill in thoroughly the attached technical data schedule for each offered item. Any offer does not accompany with clear and complete technical data schedule shall be rejected.
- Product type test reports/certificates.
- Approval certificate issued by Egyptian Electricity Holding Company.
- List of deviations and clauses to which exception is taken (if any).

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10. DRAWINGS

Aluminum Lug



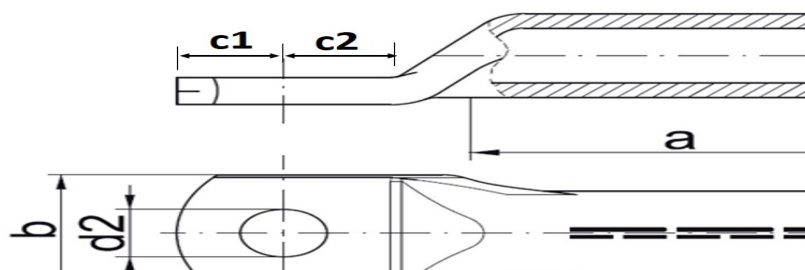
Conductor Cross-section mm ²	Stud dia.	a ^(*) Min.	b ±2	c1 ±1	c2 ±1	d1 ^(**) ±0.6	d2 ±0.5	d4 ±0.6	L ±3	Weight g/pc without inhibitor	No. of Press (crimps) narr/wide
	All dimensions are in mm										
10	8	32	18	9.5	12	5	8.5	10	52	11	4 / 2
16	8	32	18	9.5	12	6.5	8.5	12	52	12	4 / 2
25	8	38	18	9.5	12	6.8	8.5	12	60	12.6	4 / 2
35	10	42	21	12	14	8	10.5	14	67	20.8	5 / 2
50	10	42	25	14	14.5	9.8	10.5	16	72	25.4	5 / 2
70	12	52	28	15	17.5	11.2	13	18	86	40	6 / 3
95	12	52	32	16	18	13.2	13	22	90	65	6 / 3
120	12	52	32	16	18	14.7	13	22	91	55	6 / 3
150	12	60	35	17.5	21.5	16.3	13	25	103	88	6 / 3
185	10	60	40	21.5	25	18.3	11	28	106	110	6 / 3
185	16	60	40	21.5	25	18.3	17	28	106	110	6 / 3
240	12	65	45	24	26	21	13	32	116	157	6 / 3
240	16	65	45	24	26	21	17	32	116	157	6 / 3
300	16	75	49	24	26	23.3	17	34	124	178.5	8 / 3
400	20	100	58	30.5	32	26	21	38	165	290	- / 4
500	20	100	62	31	32	29	21	44	165	387	- / 4
630	20	100	62	31	32	32	21	44	165	357	- / 4
800	20	110	78	39	27	36	21	52	185	597	- / 4
1000	20	110	80	39	27	41.5	21	58	185	725	- / 4

(*) Dimension "a" denotes the length of the conductor part inside the lug barrel.

(**) For M.V termination of cables sizes 150, 240, 300, and 400mm² inner diameter "d1" accepted to reduce by 1mm, in-additional of applying tolerances in AM table.

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Copper Lug



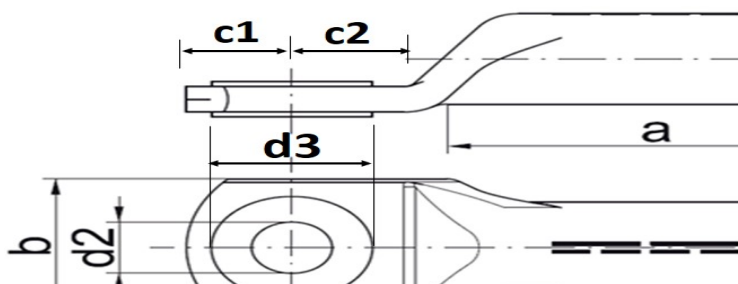
Conductor Cross-section mm ²	Stud dia.	a (*) Min.	b	c1	c2	d1	d2	d4	L	Weight g/pc	No. of Press (crimps) narr/wide
All dimensions are in mm (**)											
10	6	10	9	7.5	8.5	4.5	6.4	6	27	3.6	1 / -
16	8	20	13	10	10	5.5	8.4	8.5	36	12.2	2 / -
25	8	20	16	10	10	7	8.4	10	38	15	2 / -
35	10	20	19	12	12	8.2	10.5	12.5	42	26	2 / -
50	10	28	22	12	12	10	10.5	14.5	52	44	3 / -
70	12	28	24	12	12	11.5	13	16.5	55	61	3 / -
95	12	35	28	13	13	13.5	13	19	65	86	4 / 2
120	12	35	32	16	13	15.5	13	21	70	113	4 / 2
150	12	35	34	16	17	17	13	23.5	78	163	4 / 2
185	16	40	37	19	20	19	17	25.5	82	187.5	4 / 2
240	16	40	42	19	20	21.5	17	29	92	273	- / 2
300	16	50	46	19	22	24.5	17	32	100	333	- / 2
400	20	70	54	25	22	27.5	21	38.5	115	654	- / 3
500	20	70	60	25	22	31	21	42	125	793	- / 3
630	20	80	64	25	20	34.5	21	44	135	790	- / 3
800	20	100	75	25.3	25.3	38.4	21	52	165	1610	- / 3
1000	20	100	85	25	20	44	21	58	165	1814	- / 3

(*) Dimension "a" denotes the length of the conductor part inside the lug barrel.

(**) Tolerances are according to DIN 46235.

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Bimetallic (Al – Cu) Lug (Insert type)



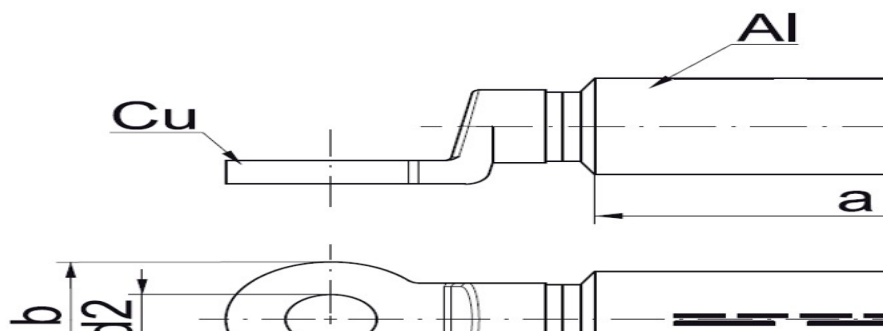
Conductor Cross-section mm ²	Stud dia.	a (*) Min.	b ±2	c1 ±1	c2 ±1	d1(**) ±0.6	d2 ±0.5	d3 Min.	d4 ±0.6	L ±3	Weight g/pc without inhibitor	No. of Press (crimps) narr/wide
All dimensions are in mm												
16	8	32	22.5	13.6	15.6	6.5	8.5	17.5	12	52	18	4 / 2
25	8	38	22.5	13.6	15.6	6.8	8.5	17.5	12	61	20	4 / 2
35	10	42	25	14	16	8	10.5	19.5	14	67.5	30	5 / 3
50	10	42	26	14	16	9.8	10.5	19.5	16	72	33.6	5 / 3
70	12	52	31.5	17	17.5	11.2	13	24.5	18	86	60.5	6 / 3
95	12	52	33.5	17.5	17	13.2	13	26	22	90	88	6 / 3
120	12	52	35	17.5	17	14.7	13	26	22	90	78	6 / 3
150	12	60	37	18.8	21.5	16.3	13	26	25	103	112	6 / 3
185	10	60	42.5	22.3	25	18.3	11	21	28	107	137.5	6 / 3
185	12	60	42.5	22.3	25	18.3	13	28	28	107	137.5	6 / 3
240	12	65	48	25	26	21	13	28	32	116	195	6 / 3
240	16	65	48	25	26	21	17	33	32	116	195	6 / 3
300	16	75	50.5	25	26	23.3	17	34	34	124	218	8 / 3
400	16	100	56	31.3	32	26	17	35	38	165	333	- / 4
500	16	100	63	30.5	32	29	17	35	44	165	462	- / 4
630	16	100	63	30.5	32	32	17	35	44	165	424	- / 4
800	20	110	78	39	27	36	21	42	52	185	642	- / 4
1000	20	110	80	39	27	41.5	21	42	58	185	800	- / 4

(*) Dimension “a” denotes the length of the conductor part inside the lug barrel.

(**) For M.V termination of cables sizes 150, 240, 300, and 400mm² inner diameter “d1” accepted to reduce by 1mm, in-additional of applying tolerances in AM table.

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Bimetallic (Al – Cu) Lug (Friction type)

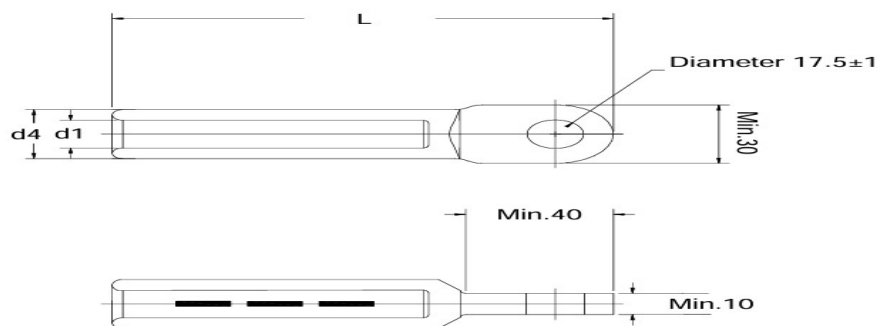


Conductor Cross-section mm ²	Stud dia.	a ^(*) Min.	b ±2	d1 ±0.6	d2 ±0.5	d4 ±0.6	L ±3	Weight g/pc without inhibitor	No. of Press (crimps) narr/wide
	All dimensions are in mm								
16	10	43	25	6.5	10.5	20	72.5	77	4 / 2
25	10	43	25	6.8	10.5	20	72.5	77	4 / 2
35	10	43	25	8	10.5	20	72.5	75	4 / 2
50	12	43	25	10	13	20	72.5	69	4 / 2
70	12	43	25	11.2	13	20	72.5	68	4 / 2
95	12	43	25	13.2	13	20	72.5	66	4 / 2
120	12	59	30	14.7	13	25	97	136	5 / 2
150	12	59	30	16.3	13	25	97	124	5 / 2
185	10	59	35	18.5	11	32	100	224	5 / 2
185	12	59	35	18.5	13	32	100	224	5 / 2
240	16	59	35	21	17	32	100	203	5 / 2
300	16	73	36	23.3	17	34	128	252	- / 4
400	16	73	36	26	17	38.5	128	292	- / 4
500	16	80	36	29	17	44	135	292	- / 4
630	16	94	60*60	32	17	44	200	855	- / 4

(*) Dimension “a” denotes the length of the conductor part inside the lug barrel.

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Tin plated Alumium Lugs for separeble compression type



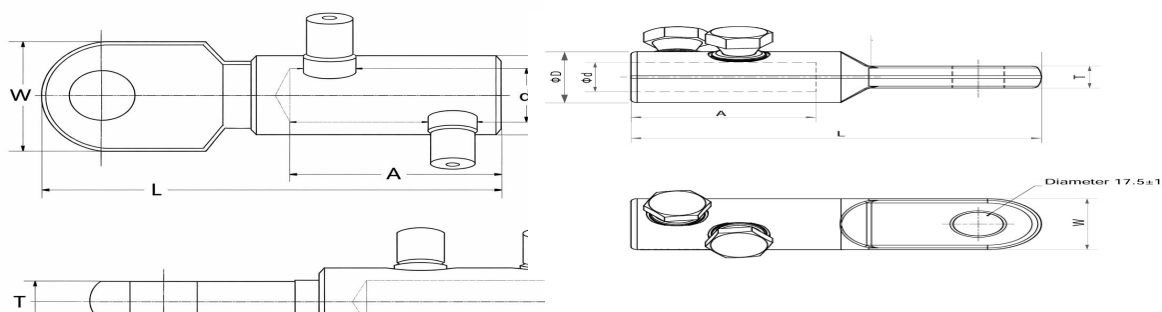
TINNED AL LUGS				
(For dead break elbow of bushing interface B, D or C up to 630A suitable for AL & CU cables)				
Conductor Cross-section mm²	d1^(*)	d4^(*)	Min. L	Crimping die size
All dimensions are in mm				
70	11.2	22	120	22
95	12.3	22	120	22
120	14	22	120	22
185	18.3	28.5	120	28
240	20.2	32.5	130	32
300	22.7	32.5	130	32
400	25.5	42	130	42
500	28.6	44	130	44
630	32.5	44.5	130	44

TINNED CU LUGS				
(For dead break elbow of bushing interface C 1250A suitable for CU cables)				
500	31	42	150	42
630	34.5	44	150	44

- (*) The above dimensions are subjected to a manufacturing tolerance up to ± 1 mm.
- Length of lugs can be increase according to manufacturer design.
- Minimum number of compressions charts are 3.

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Tin plated Aluminum Mechanical Lugs for separable connectors



Conductor size (mm) ²	Min. Dia d (mm)	Min. Dia D (mm)	Min. A (mm)	Min. L (mm)	Min. T (mm)	Min. W (mm)	Min. No. of Bolts
70	13	25	30	84.5	10	30	1
95	15.5	29	35	96	10	30	1
120	15.5	29	35	96	10	30	1
185	20	33	56	116	12.5	35	2
240	20	33	56	116	12.5	35	2
300	23	36	67	125	12.5	35	2
400	25	42	75	130	12.7	36	3
500	33	50	90	155	12.7	36	3
630	33	52	90	155	12.7	36	3

- The above dimensions are subjected to a manufacturing tolerance.
- Length of lugs and palm diameter can be increased according to the manufacturer's design.

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11. For TECHNICAL DATA SCHEDULE

__ EDC Inquiry No: _____ Item No: _____

No.	DESCRIPTION	__ EDC Specified Values	Vendor Proposed Values
1	Lugs type	share bolt / compression	
2	Manufacturer	---	
3	Manufacturer Catalogue No.	---	
4	Country of Origin	---	
5	Material	Cu / Al / Bimetallic/ Tin plated AL	
6	Material Compliance	Yes / No	
7	Material purity (%)	Cu 99.9 % / Al 99.5 %	
8	Min. Material conductivity (IACS)	Cu 96 % / Al 58 % (comp. type) (36% for shear bolt type)	
9	Cable / Conductor Size (mm ²)	10 up to 1000	
10	Bolt Size	M6 / M8 / M10 / M12 / M16 / M20	
11	Min.Tin-Plating Thickness (μm)	15 (for cooper & dead break elbow lugs)	
12	Submittals Required Included	Yes / No	

I / We guarantee the technical data given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		